

Exercise 02: Runtime Function

Write a text file named 'runtime.txt' that calculates the runtime function of each program below by

1. stating its input size
2. constructing its runtime table (or list)

Programs

1. Program 01

```
Prog01(A)
01 | s ← 0
02 | for i = 0 to A.length, do
03 |   s ← s + A[i]
04 | return s / A.length
```

2. Program 02

```
Prog02(A)
01 | for i = 0 to A.length / 2, do
02 |   t ← A[i]
03 |   A[i] ← A[A.length-1-i]
04 |   A[A.length-1-i] ← t
```

3. Program 03

```
Prog03(A)
01 | for i = 0 to A.length, do
02 |   for j = i to A.length, do
03 |     if A[i] = A[j], then
04 |       return false
05 | return true
```